The following formulas may be used
$x_{\text{new}} = x_{\text{current}} + \beta \cdot \delta \cdot \text{sign}\left(f(x_{\text{current}}) - f\left(x_{\text{previous}}\right)\right)$
$v_i(t+1) = w \times v_i(t) + c_1 \times r_1 \times (pbest_i - x_i(t)) + c_2 \times r_2 \times (gbest_i - x_i(t))$
$p_k(i,j) = \begin{cases} \frac{[\tau_{ij}]^{\alpha}[\eta_{ij}]^{\beta}}{\sum_{u \in N_K(i)}[\tau_{iu}]^{\alpha}[\eta_{iu}]^{\beta}}, & if\ j \in N_K(i) \\ 0 & otherwise \end{cases}$
$v_i^t = v_i^{t-1} + (x_i^{t-1} - x_*) \cdot f_i$
$r_i^{t+1} = r_i^0[1 - e^{-\gamma t}]$
$\sigma^2 = \left[\frac{\Gamma(1+\lambda)}{\lambda\Gamma((1+\lambda)/2)} \cdot \frac{sin(\pi\lambda/2)}{2^{(\lambda-1)/2}}\right]^{1/\lambda}$
$x_{ij} = x_j^{min} + rand(0,1) \cdot \left(x_j^{max} - x_j^{min}\right)$
$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \left\lvert \vec{C} \cdot \vec{X}_p(t) - \vec{X}(t) \right\rvert$
$\vec{X}(t+1) = \overrightarrow{D'} e^{bl} \cos(2\pi l) + \overrightarrow{X^*}(t)$
$x_i(t+1) = x_i + \beta_0 e^{-\gamma r^2{}_{i,j}}\left(x_j - x_i\right) + \alpha\varepsilon_i$
$\boldsymbol{x}_i^{t+1} = \boldsymbol{x}_i^t + \gamma L(\lambda)(\boldsymbol{g}_* - \boldsymbol{x}_i^t)$ $x_i^{t+1} = x_i^t + \epsilon\left(x_j^t - x_k^t\right),$
$L \sim \frac{\lambda\Gamma(\lambda)\sin(\pi\lambda/2)}{\pi} \frac{1}{s^{1+\lambda}},\ (s \gg s_0 > 0).$
$s = \frac{U}{\lvert V \rvert^{1/\lambda}},$
$v_{ij} = x_{ij} + \phi_{ij}\left(x_{ij} - x_{kj}\right)$
$v_i = x_{r1} + F \times (x_{r2} - x_{r3})$
$u_{i,j} = \begin{cases} v_{i,j} & if\ rand_j < CR \\ x_{i,j} & Otherwie \end{cases}$